

Advancing Legal Reasoning in *Algerian Law*

RAG · KNOWLEDGE GRAPHS · ARGUMENT MINING → *Citation-faithful legal answers*Ibrahim El Khalil Attia · Maab Chaoui | Supervisor **Dr. Fouad Dahak** Co-supervisor **Prof. Ivan Cantador**

01 The Problem

WHY WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.

SOURCED · ARTICLE-CITED · JURISDICTION-BOUND ·
AMENDMENT-AWARE · OR *REFUSED*

1.2 What legal questions demand

بين الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي،
وحدود سلطته عند غياب النص...

— demands exact articles — *Civil Code, arts. 1–2*

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل
؟2005

— demands the law as amended — *Ord. 05-02 → Family Code, art. 54*

هل يحق لصاحب العمل في الجزائر فصل العالِم
تامة دون إبداء أسباب (at-will) ؟

— a foreign-doctrine trap — *Algerian law is silent; the system must abstain*verbatim from *AlgerianLegalBench* (244 questions)

1.3 Where today's LLMs fail

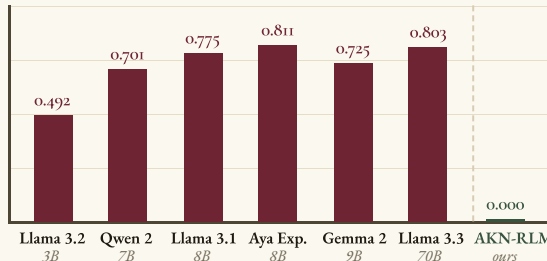


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not exist*, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. AKN-RLM: 0.000.

1.4 ...and Algerian law raises the bar

- **Arabic-first.** 232 of 244 benchmark questions are in Arabic
- **Layered in time.** 1963–2025; amendments rewrite articles in place
- **Deeply structured.** code → chapter → section → article
- **Low-resource.** barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ **Structure the law**
171 docs → *Akoma Ntoso XML* + legal KG
- ✓ **Benchmark it**
AlgerianLegalBench: 244 expert questions
- ✓ **Reason on evidence**
typed retrieval + bounded recursion
- ✓ **Verify or abstain**
triple faithfulness gate

1.6 Significance & impact

1. **First citation-faithful QA system for Algerian law** — every claim resolves to a registered article, or the system abstains.
2. **Reusable infrastructure for Algerian legal AI**
 - **AKN corpus + knowledge graph** — 171 structured documents, 1963–2025, article-level registry
 - **AlgerianLegalBench** — 244 expert-annotated questions, $\kappa = 0.829$
3. **A recipe for low-resource legal AI** — evidence control beats parameter scale.

02 The Solution

HOW STRUCTURED CORPUS → BENCHMARK → AKN-RLM

I Structured legal corpus

the law, made machine-readable

What is Akoma Ntoso? The OASIS LegalDocML standard — an XML vocabulary for legal texts in which every law, article, amendment and cross-reference becomes an **addressable node** under a canonical FRBR URI: the article registry that every citation must resolve against.

WHAT WE BUILT

171 source files
189 MB

45 laws & codes
→ AKN XML

46 RDF graphs
→ one legal KG

1963–2025 coverage,
amendments
in place

```
<!-- Civil Code – Ordinance 75-58, as amended -->
<act name="code_civil">
  <FRBRWork value="/akn/dz/act/1975-09-26/75-58"/>
  <FRBRExpression value=".../ara@2005-06-20"/>
  <lifecycle>
    <eventRef date="2005-06-20" type="amendment"
      source="#law-05-10"/>
  </lifecycle>
  <article eId="art_124">
    <num>124</num> المادة</num>
    <content>كل فعل أيا كان يرتكبه الشخص بخطئه</content>
    <ref href="#art_127"/>
  </article>
</act>
```



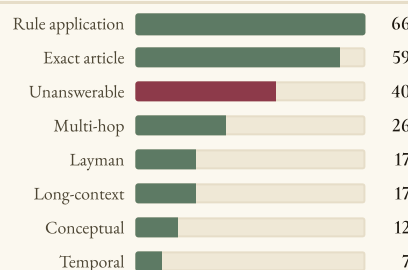
Validated in **multiple manual passes** by two law students and a supervising senior legal expert — the structure is derived from the sources, never invented.

II AlgerianLegalBench

the evaluation layer

244 expert-annotated questions across 23 legal categories — 177 answerable, 67 expecting **abstention** · 232 Arabic / 12 French · each carries expected articles, a reference answer and a **four-step reasoning chain** · double-annotated, $\kappa = 0.829$.

QUERY TYPES



DIFFICULTY



III AKN-RLM

retrieve → reason → verify

1 Retrieve evidence

RRF(BM25 + Dense)



Legal question
AR · FR

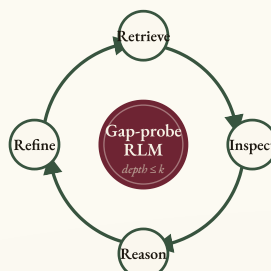


Classifier
routes 8 types



Hybrid retrieval
typed handlers

2 Reason recursively

bounded · gap-driven
depth-limited

- Detects evidence gaps
insufficient support → expand
- Toulmin ADU mining
on every cited article
- One corrective retry
then stops — no open recursion

✓ Evidence sufficient → verify & answer

3 Verify & answer

triple faithfulness gate



Citation exists
vs registry



Jurisdiction OK
canary check



Claim supported
multiling. NLI



Verifiable answer
every claim → a registered article



Principled abstention
when the law is silent

03 Results

PROOF FROZEN 244-QUESTION RUN

0.305 Citation F1
locked deployable

✓ 0.000
HCR · JIR✓ 0.703
Abstention F1

1.64×

vs direct LLM

2.9×

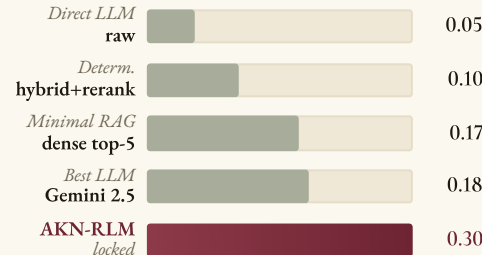
vs determ. retr.

1.74×

vs minimal RAG

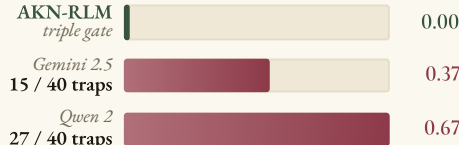
▬ Citation F1 vs baselines

HIGHER = BETTER



✓ Jurisdictional infection (JIR)

LOWER = SAFER



Only AKN-RLM holds **0.000** hallucinated-citation & foreign-law rates. Reliability tracks **legal structure, not scale** — the largest baselines hallucinate most.

⚠ LIMITS

- Retrieval ceiling: R@10 = 0.258
- French under-served
- Abstention precision 0.654
- Metrics are proxies

🕒 NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review



Structured sources

+



Controlled retrieval

+



Recursive reasoning

+



Citation verification

=

THE TAKEAWAY

Trustworthy Legal AI

an evidence-controlled system, not an unconstrained generator.